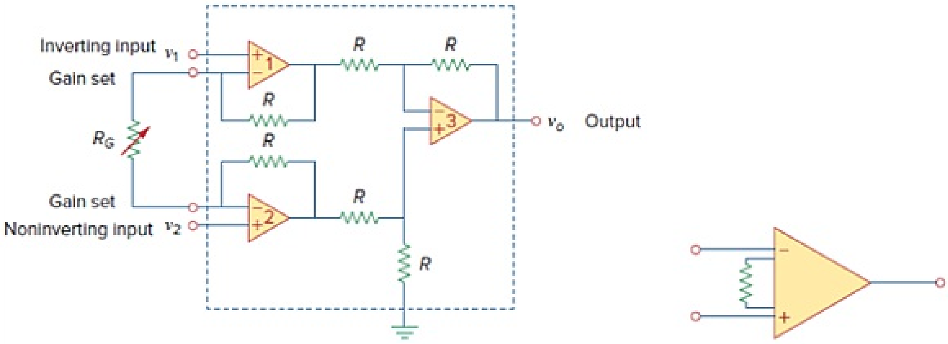
Chapter 5, Problem 13PP

Problem

Determine the value of the external gain-setting resistor *RG* required for the IA in Fig. 5.38 to produce a gain of 200 when *R* = 25 kΩ.

Figure. 5.38: (a) The instrumentation amplifier with an external resistance to adjust the gain, (b) schematic diagram.



(a) (b)

Step-by-step solution

Step 1 of 1

Given produce gain $A_V = 142$

The given resistance $R = 25\ \text{k}\Omega$

The voltage gain we have is

$$A_V = 1 + \frac{2R}{R_G}$$

$$142 = 1 + \frac{2 \times 25\text{k}}{R_G}$$

$$142 = 1 + \frac{50\text{k}}{R_G}$$

$$\frac{50}{R_G} = 142 - 1$$

$$\frac{50}{R_G} = 141$$

$$R_G = \frac{50 \times 10^3}{141}$$

$$\boxed{R_G = 354.6\Omega}$$